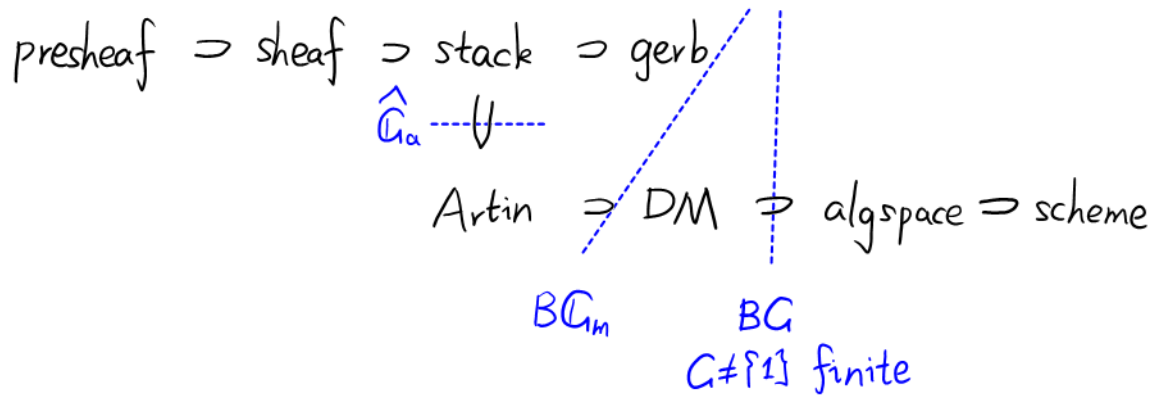


Eine Woche, ein Beispiel
 5.18 gerb & twisted sheaves

This document records the learning from AI. Use at your own risk!



$$\begin{aligned}
 \hat{G}_a &= \text{Spf } k[[t]] & \longrightarrow & \text{Spec } k[[t]] \\
 &= \varinjlim_n \text{Spec } k[t]/(t^n) & & \\
 &= \{ (t) \} & & = \{ (0), (t) \}
 \end{aligned}$$

Def. (stack in groupoids) Let $\mathcal{F} \in Sh(X)$.

$\mathcal{F}: Open(X)^{\text{op}} \longrightarrow \text{Gpd}$ is a stack, if

$$\mathcal{F}(\mathcal{U}) \xrightarrow{\sim} \text{holim} \left(\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_{ij}) \rightrightarrows \prod_{i,j,k} \mathcal{F}(U_{ijk}) \right)$$

as an equivalence of groupoids, where

$$\begin{aligned} Ob(\text{holim}(\dots)) &= \{ \text{descent datum } (\mathcal{L}_i, \phi_{ij}) \} \\ &= \left\{ (\mathcal{L}_i, \phi_{ij})_{i,j} \mid \begin{array}{l} \mathcal{L}_i \in \mathcal{F}(U_i) \\ \phi_{ij}: \mathcal{L}_i|_{U_{ij}} \xrightarrow{\sim} \mathcal{L}_j|_{U_{ij}} \\ \phi_{jk} \circ \phi_{ij}|_{U_{ijk}} = \phi_{ik}|_{U_{ijk}} \end{array} \right\} \end{aligned}$$

$$\text{Mor}((\mathcal{L}_i, \phi_{ij}), (\mathcal{L}'_i, \phi'_{ij})) = \left\{ (f_i)_i \mid \begin{array}{ccc} f_i: \mathcal{L}_i & \xrightarrow{\sim} & \mathcal{L}'_i \\ \mathcal{L}_i|_{U_{ij}} & \xrightarrow{f_i|_{U_{ij}}} & \mathcal{L}'_i|_{U_{ij}} \\ \phi_{ij} \downarrow & \circlearrowright & \downarrow \phi'_{ij} \\ \mathcal{L}_j|_{U_{ij}} & \xrightarrow{f_j|_{U_{ij}}} & \mathcal{L}'_j|_{U_{ij}} \end{array} \right\}$$

E.g. $Sh^{\sim} \supset Q\text{Coh}^{\sim} \supset \text{Coh}^{\sim} \supset B\mathbb{G}_m$
 stack of sheaves
 are stacks, where

$$\begin{aligned} Ob(Sh^{\sim}(\mathcal{U})) &= \{ \text{sheaves on } \mathcal{U} \} & = Ob(Sh(\mathcal{U})) \\ \text{Mor}_{Sh^{\sim}(\mathcal{U})}(\mathcal{F}, \mathcal{F}') &= \{ \text{Iso of sheaves on } \mathcal{U} \} & \subseteq \text{Mor}_{Sh(\mathcal{U})}(\mathcal{F}, \mathcal{F}') \end{aligned}$$

$$\begin{aligned} Ob(B\mathbb{G}_m(\mathcal{U})) &= \text{Pic}(\mathcal{U}) \\ \text{Mor}_{B\mathbb{G}_m(\mathcal{U})}(\mathcal{L}, \mathcal{L}') &= \text{Isom}_{\mathcal{O}_{\mathcal{U}}}(\mathcal{L}, \mathcal{L}') \end{aligned}$$

e.g. $\text{Mor}_{B\mathbb{G}_m(\mathcal{U})}(\mathcal{L}, \mathcal{L}) = \mathbb{G}_m(\mathcal{U})$

Def (Gerby) A stack $\mathcal{G}: \text{Open}(X) \rightarrow \text{Gpd}$ is called a gerby, if (i) + (ii) + compatibility:

- (i) $\forall \mathcal{U}$, $\exists$ an open cover $\mathcal{U} = \bigcup_i \mathcal{U}_i$ s.t. $\mathcal{G}(\mathcal{U}_i) \neq \emptyset$;
(ii) $\forall \mathcal{L}, \mathcal{L}' \in \mathcal{G}(\mathcal{U})$, $\exists$ an open cover $\mathcal{U} = \bigcup_i \mathcal{U}_i$ s.t. $\mathcal{L}|_{\mathcal{U}_i} \cong \mathcal{L}'|_{\mathcal{U}_i}$.

Furthermore, for G : sheaf of gps on X , we call a stack $\mathcal{G}$ as a G -gerby, if (i) + (ii)' + compatibility:

- (ii)' $\forall \mathcal{L} \in \mathcal{G}(\mathcal{U})$, $\exists$ an open cover $\mathcal{U} = \bigcup_i \mathcal{U}_i$ s.t. $\mathcal{L}|_{\mathcal{U}_i} \cong B(G|_{\mathcal{U}_i})$;

or: $\forall \mathcal{L} \in \mathcal{G}(\mathcal{U})$, $\exists$ an open cover $\mathcal{U} = \bigcup_i \mathcal{U}_i$ s.t. $\text{Aut}_{G(\mathcal{U})}(\mathcal{L}) \cong G|_{\mathcal{U}_i}$
 $\uparrow$
sheaf of automorphisms

E.g. BG is a G -gerby over X .
 ${}^k\mathcal{I}$ is a μ_k -gerby over X , ${}^k\sqrt{\mathcal{O}_X} \cong B\mu_k$.

Here,

$$\begin{aligned} G &: \text{sheaf of gps on } X \\ \text{Ob}(BG(\mathcal{U})) &= \{G|_{\mathcal{U}}\text{-torsors on } \mathcal{U}\} \\ \text{Mor}(\mathcal{L}, \mathcal{L}') &= \{\varphi: \mathcal{L} \xrightarrow{\sim} \mathcal{L}' \text{ is } G|_{\mathcal{U}}\text{-equivalent}\} \\ \text{e.p. } \text{Mor}(G|_{\mathcal{U}}, G|_{\mathcal{U}}) &= G(\mathcal{U}) \end{aligned}$$

$$\begin{aligned} \text{Ob}({}^k\mathcal{I}(\mathcal{U})) &= \left\{ (M, \varphi) \mid \begin{array}{l} M \in \text{Pic}(\mathcal{U}) \\ \varphi: M^{\otimes k} \rightarrow \mathcal{L}|_{\mathcal{U}} \end{array} \right\} \\ \text{Mor}((M, \varphi), (M', \varphi')) &= \left\{ \begin{array}{l} \phi: M \rightarrow M' \text{ s.t.} \\ \begin{array}{ccc} M^{\otimes k} & \xrightarrow{\phi^{\otimes k}} & M'^{\otimes k} \\ \varphi \downarrow & \circlearrowleft & \downarrow \varphi' \\ & \mathcal{L}|_{\mathcal{U}} & \end{array} \end{array} \right\} \end{aligned}$$

Def (The site of objects)

For $X \in \mathbf{Psh}(X)$, the site of objects $\int X$ is a category where

$$\begin{aligned} \text{Ob}(\int X) &= \{(u, x) \mid u \in \text{Open}(X), x \in X(u)\} \\ \text{Mor}((v, y), (u, x)) &= \left\{ (f, \phi) \mid \begin{array}{l} f: v \hookrightarrow u \\ \phi \in \text{Mor}_{X(v)}(x|_v, y), \phi: x|_v \xrightarrow{\sim} y \end{array} \right\} \end{aligned}$$

The 2-functor

$$x|_v = f^*_x$$

$$\int: \text{Fun}(X^{\text{op}}, \text{Set}) \longrightarrow \text{Cat}$$

is called the Grothendieck construction.

Rmk. $\int X$ carries a natural Grothendieck topo.
i.e., $\int X$ is a site.

$$\begin{aligned} \mathcal{T}: \int X &\longrightarrow \bigcup_{(u, x) \in \int X} \mathcal{P}(\{\text{sieves on } (u, x)\}) \\ (u, x) &\longmapsto \mathcal{T}((u, x)) := \left\{ \{ (u_i, x_i) \}_i \mid \bigcup_i u_i = u \right\} \end{aligned}$$

Def. (sheaf on presheaf)

Let $X \in \text{Psh}(X)$. A sheaf $\mathcal{F}$ on X is a functor

$$\mathcal{F}: (\int X)^{\text{op}} \longrightarrow \text{Set}$$

satisfying the sheaf condition:

$\forall$ open cover $\{U_i \hookrightarrow U\}$ and $\forall x \in X(U)$,

$$\mathcal{F}(U, x) \longrightarrow \prod_i \mathcal{F}(U_i, x|_{U_i}) \rightrightarrows \prod_{i,j} \mathcal{F}(U_{ij}, x|_{U_{ij}})$$

is an equalizer.

E.g. When $X = h_X: \text{Open}(X) \longrightarrow \text{Set}$
 $U \longmapsto \{*\} = \{U \hookrightarrow X\}$
 we get $\int h_X = \text{Open}(X)$, $\text{Sh}(h_X) = \text{Sh}(X)$.

When $X = \mathcal{F}$ is a sheaf,

$$\begin{aligned} \int \mathcal{F} &= \{(U, x) \mid U \in \text{Open}(X), x \in \mathcal{F}(U)\} \\ &= \text{"category of local sections on } X\text{"} \\ &= \text{Sect}(\underbrace{V(\mathcal{F})}_{\uparrow \text{ sections}}, X) \end{aligned}$$

Def (twisted sheaf)
 Suppose $G \in \mathbb{G}_m\text{-Gerby}_X$.

A G -twisted sheaf is a sheaf $\mathcal{F}$ on X
 s.t. $\forall p \in X, \exists U \ni p$ open s.t.

$$\text{Aut}(G|_U) = \mathbb{G}_m \hookrightarrow \mathcal{F}|_U$$

by standard scalar action.

$$\text{Coh}(X, \alpha) = \text{Coh}^{\mathbb{G}_m\text{-equiv, wt } 1}(\mathcal{G}_\alpha)$$

Rmk. For $G \in \text{Sh}(X)$ a sheaf of abelian gps, we have

$$\begin{array}{c} \{G\text{-banded gerbs on } X\} / \text{equiv} \cong H^2(X; G) \\ \uparrow \\ \text{"canonical identified"} \end{array}$$

E.g. $\sqrt[k]{\mathcal{L}} \in H^2(X; \mu_k)$ is constructed via SES:

$$1 \longrightarrow \mu_k \longrightarrow \mathbb{G}_m \xrightarrow{(-)^k} \mathbb{G}_m \longrightarrow 1$$

$$\hookrightarrow H^2(X; \mu_k) \longrightarrow H^2(X; \mathbb{G}_m)$$

$$\begin{array}{ccc} H^1(X; \mathbb{G}_m) & \longrightarrow & H^1(X; \mathbb{G}_m) \\ \mathcal{L} & \longmapsto & \mathcal{L}^{\otimes k} \end{array} \quad \mathcal{L} \mapsto [\sqrt[k]{\mathcal{L}}]$$